

Harshitha Pothini
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Detail-oriented Data Analyst and Machine Learning enthusiast with hands-on experience in Python, SQL, Power BI, and Excel, specializing in predictive analytics, spatial data integration, and interactive dashboard development. Proven ability to transform raw data into actionable insights across diverse domains, including climate science, retail, and higher education. At Southern Illinois University Edwardsville, I contributed to a \$15 million fundraising campaign by managing large datasets, ensuring data accuracy, and enabling targeted donor outreach through complex SQL queries, automated reporting workflows, and data visualization. Skilled in deep learning models such as LSTM, statistical analysis, and Agile methodologies, with a strong track record of collaborating with cross-functional teams to align analytics with strategic business goals and drive operational efficiency.

Work Experience:

Data Analyst (Graduate Assistant)

November 2024 to Present

Southern Illinois University Edwardsville – Edwardsville

- Analyzed and managed large datasets of university student and donor records, ensuring data accuracy and integrity to support institutional reporting and strategic decision-making.
- Developed and maintained Power BI dashboards to visualize key operational and fundraising metrics, improving leadership insight and data-driven planning.
- Utilized SQL queries to gather and manipulate data from SQL Server, enabling precise extraction for reporting and analysis needs.
- Produced numerous ad hoc reports and conducted in-depth data validation to ensure accuracy and support strategic goals, including a \$15 million fundraising target for the current fiscal year.
- Maintained and updated constituent records, including accurate identification and management of deceased records, supporting data hygiene and compliance.
- Updated constituent records, including identifying and maintaining deceased records, to enhance data quality and compliance.
- Manage exports and conduct weekly queries for donor postcards.
- Extracted, manipulated, and integrated data from multiple sources using SQL for accurate transformation and analysis.
- Participated in Agile workflows, including daily stand-ups and sprint planning, to ensure team collaboration and project success.
- Collaborated regularly with the Director of Operations and other stakeholders to align data strategy with business objectives and ensure accurate delivery of actionable insights.
- Managed data exports and conducted weekly queries for targeted donor outreach, such as personalized postcard campaigns.
- Utilized advanced Excel functions, pivot tables, and automation techniques to streamline reporting processes, identify key trends, reduce manual errors, and generate actionable business insights.

Data Analyst

September 2023 to March 2024

Cognizant Technologies- Kolkata, India

- Transformed large-scale transactional and customer data into actionable insights using SQL, Power BI, and Excel, supporting merchandising, marketing, and store operations teams.

- Developed interactive Power BI dashboards to track retail KPIs including in-store and online sales, product performance, customer segmentation, and promotional impact.
- Collaborated with cross-functional teams at Macy's to design and implement data models and DAX measures, improving reporting accuracy and decision-making speed.
- Led ETL processes in collaboration with the data engineering team, cleaning and normalizing sales and inventory data across departments to ensure consistent, high-quality reporting.
- Python and R applied to perform customer behavior analysis, sales forecasting, enabling targeted marketing campaigns and loyalty initiatives.
- Automated daily and weekly reporting workflows using Power Query and advanced Excel functions (VLOOKUP, INDEX-MATCH, pivot tables), reducing manual effort by 50%.
- Participated in Agile sprints, working closely with product owners and data engineers to deliver analytics features aligned with Macy's seasonal and campaign cycles.
- Conducted in-depth product category analysis, helping buyers identify underperforming SKUs and optimize pricing and markdown strategies.
- Ensured data integrity by validating order and customer records across multiple platforms (ERP, CRM, POS), resolving inconsistencies that impacted revenue reporting.
- Delivered weekly executive reports and visual summaries highlighting regional sales trends, return rates, and customer engagement metrics to senior leadership.
- Supported Black Friday and seasonal campaign readiness by building high-frequency dashboards and anomaly detection models to monitor real-time performance.

Data Analyst Intern

February 2023 to May 2023

Cognizant Technologies -Hyderabad, India

- Assisted in preparing Power BI reports and dashboards under the guidance of senior analysts to support client-facing insights.
- Supported ML-based dashboard development using Power BI and Python for internal analytics projects.
- Wrote and optimized basic SQL queries to extract data for reports and ad hoc analysis.
- Supported the data team in cleaning, transforming, and validating data using Excel and Power Query.
- Learned and applied key ETL concepts, helping to prepare raw data for visualization and reporting.
- Built initial prototypes for classification models and conducted EDA on retail datasets using Pandas and Matplotlib.
- Learned and applied ML model evaluation and feature selection methods under mentorship.
- Participated in team stand-up meetings and sprint planning as part of the Agile delivery process.
- Helped with testing and documentation of dashboards, ensuring end-users received accurate and user-friendly reports.
- Shadowed senior BI consultants to understand real-world retail data use cases and stakeholder interactions.

Data Analyst Intern

September 2022 to January 2023

Indian Servers

- Assisted in developing Excel dashboards to visualize performance metrics for internal projects and client demos.
- Developed an image classification model using machine learning techniques in Python, applying libraries like TensorFlow/Keras.
- Collected and preprocessed image datasets, applied CNN architectures, and achieved high classification accuracy.

- Gained hands-on experience with model training, hyperparameter tuning, and result visualization.
- Collaborated with the analytics team to support ML-focused internal projects and research.
- Collaborated with the analytics team to support small-scale projects focused on customer feedback and service optimization.
- Conducted exploratory data analysis (EDA) to identify trends and outliers in operational datasets.
- Documented project steps and findings to support future reuse and knowledge sharing.
- Gained hands-on experience with real-world datasets and learned to translate business questions into data-driven insights

Education:

Master's degree in computer science
Southern Illinois University - Edwardsville, Illinois

January 2024 to December 2025

Bachelor's degree in computer science
RVR&JC - Andhra Pradesh

August 2019 to May 2023

Projects in Graduate School

- **Analyzing the Influence of Climate Variables on Hurricane Intensity**

Conducted an advanced research project using LSTM neural networks to predict hurricane intensity from climate data including CO₂, methane emissions, temperature anomalies, and precipitation. Built multivariate time series models and regression classifiers to evaluate hurricane wind speed and severity. Integrated spatial analysis using GIS to correlate emission hotspots with hurricane-prone regions. Demonstrated how climate variables, particularly greenhouse gases, influence storm behavior, aiding in climate resilience planning.

- **A Multi-Agent Approach For Hypertension Risk Prediction**

This is useful for predicting hypertension. I worked as part of a team on a project to predict hypertension risks using different data sources and machine learning methods. I helped with organizing data, creating features, and building models. Analyzed model results to improve accuracy. This project focused on understanding health data and making predictions about hypertension

Skills:

- Programming Skills: Python, SQL, C, R, Java, Machine Learning, Deep Learning, HTML, CSS, JavaScript, Spatial Analysis.
- Proficient in ETL, data purification, and constructing data models.
- Adept in using Power BI, Tableau, and Excel for insightful visualizations.
- Skilled in MySQL for database administration.
- Effective in teamwork and resolving complex issues.
- Knowledgeable in utilizing A/B testing for decision support and overseeing projects.
- Familiar with Excel for data-related tasks and Git for source control.

Certifications:

Udemy Data Analysis and Visualization Certification

- Udemy Certified Network Associate.
- Udemy – Data Analysis and Visualization
- Hacker Rank – SQL